

Q1 - 25 July - Shift 1

Which of the following physical quantities have the same dimensions ?

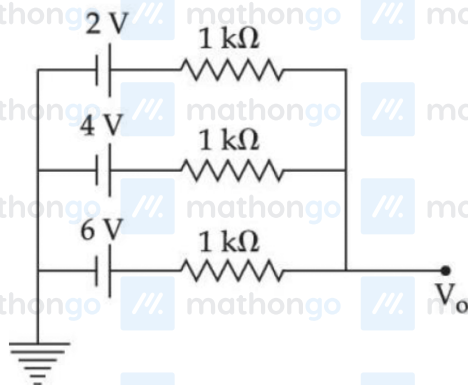
- (A) Electric displacement ($\vec{D}$) and surface charge density
- (B) Displacement current and electric field
- (C) Current density and surface charge density
- (D) Electric potential and energy

Space for your notes:

Q2 - 25 July - Shift 1

In the given figure, the value of V_0 will be _____ V.

Space for your notes:



Q3 - 25 July - Shift 1

Eight copper wire of length l and diameter d are joined in parallel to form a single composite conductor of resistance R . If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be _____ d .

Space for your notes:

Q4 - 25 July - Shift 2

Questions

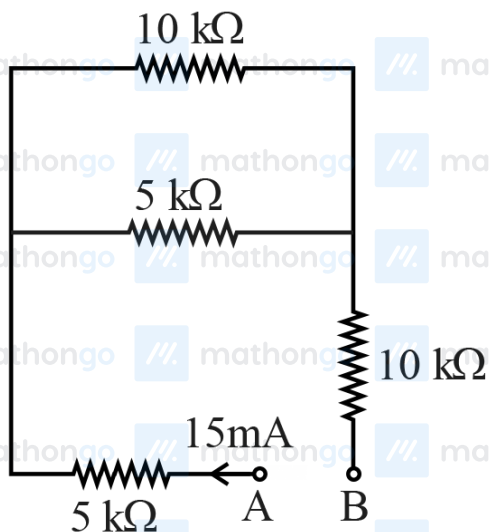
MathonGo

In AM modulation, a signal is modulated on a carrier wave such that maximum and minimum amplitude are found to be 6V and 2V respectively. The modulation index is

- (A) 100% (B) 80%
(C) 60% (D) 50%

*Space for your notes:***Q5 - 25 July - Shift 2**

A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be

Space for your notes:

- (A) 50V (B) 75V
(C) 150V (D) 275V

Q6 - 25 July - Shift 2

In a potentiometer arrangement, a cell of emf 1.20 V gives a balance point at 36 cm length of wire. This cell is now replaced by another cell of emf 1.80 V. The difference in balancing length of potentiometer wire in above conditions will be _____ cm.

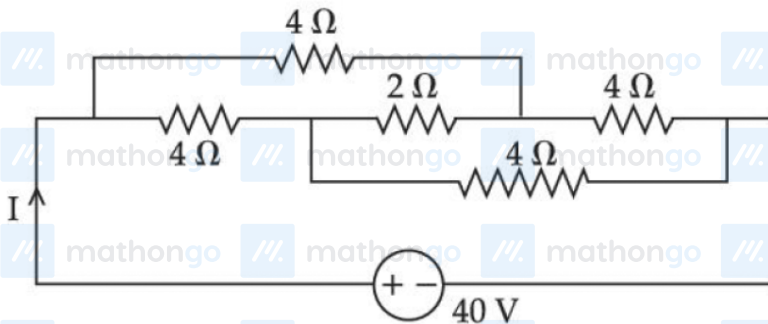
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#MathBoleTohMathonGo

Q7 - 26 July - Shift 1

The current I in the given circuit will be :

Space for your notes:



(A) 10A

(B) 20 A

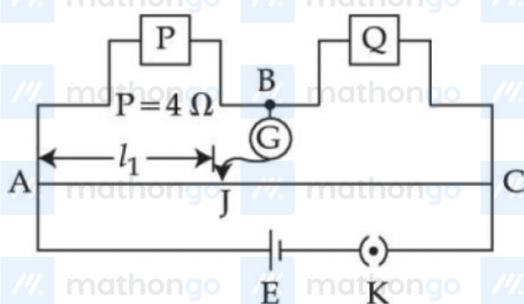
(C) 4A

(D) 40A

Q8 - 26 July - Shift 1

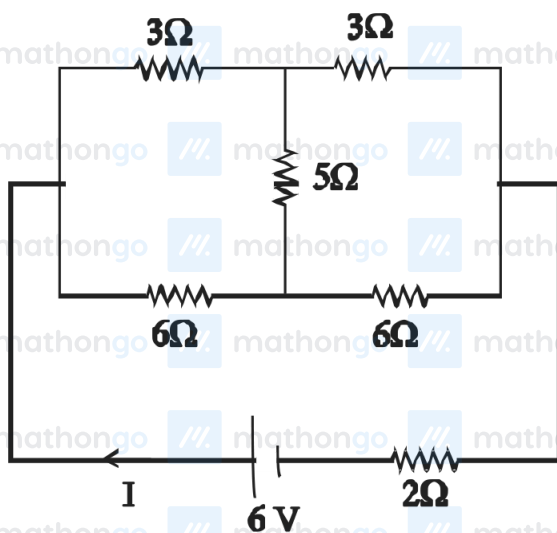
Resistance are connected in a meter bridge circuit as shown in the figure. The balancing length l_1 is 40cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80cm measured from the same end. Then the value of x will be _____ Ω

Space for your notes:



Q9 - 26 July - Shift 2

A battery of 6 V is connected to the circuit as shown below. The current I drawn from the battery is :



Space for your notes:

(A) 1A

(B) 2A

(C) $\frac{6}{11}$ A

(D) $\frac{4}{3}$ A

Q10 - 26 July - Shift 2

A potentiometer wire of length 300 cm is connected in series with a resistance $780\ \Omega$ and a standard cell of emf 4V. A constant current flows through potentiometer wire. The length of the null point for cell of emf 20 mV is found to be 60 cm. The resistance of the potentiometer wire is $_____\Omega$.

Space for your notes:

Q11 - 27 July - Shift 1

Questions

MathonGo

Two sources of equal emfs are connected in series. This combination is connected to an external resistance R . The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero then the value of R will be

- (A) $r_1 - r_2$ (B) $\frac{r_1 r_2}{r_1 + r_2}$
(C) $\frac{r_1 + r_2}{2}$ (D) $r_2 - r_1$

*Space for your notes:***Q12 - 27 July - Shift 1**

A direct current of 4 A and an alternating current of peak value 4 A flow through resistance of $3\ \Omega$ and $2\ \Omega$ respectively. The ratio of heat produced in the two resistances in same interval of time will be :

- (A) 3 : 2 (B) 3 : 1
(C) 3 : 4 (D) 4 : 3

*Space for your notes:***Q13 - 27 July - Shift 1**

A 1 m long copper wire carries a current of 1 A. If the cross section of the wire is 2.0 mm^2 and the resistivity of copper is $1.7 \times 10^{-8}\ \Omega\text{ m}$, the force experienced by moving electron in the wire is $\text{_____} \times 10^{-23}\text{ N}$. (charge on electron $= 1.6 \times 10^{-19}\text{ C}$)

Space for your notes:

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Q14 - 27 July - Shift 2

- A. The drift velocity of electrons decreases with the increase in the temperature of conductor.
- B. The drift velocity is inversely proportional to the area of cross-section of given conductor.
- C. The drift velocity does not depend on the applied potential difference to the conductor.
- D. The drift velocity of electron is inversely proportional to the length of the conductor.
- E. The drift velocity increases with the increase in the temperature of conductor.

Choose the correct answer from the options given below:

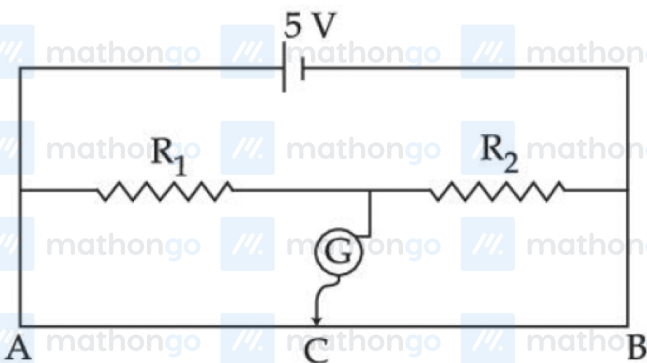
- (A) A and B only (B) A and D only
(C) B and E only (D) B and C only

Space for your notes:

Q15 - 27 July - Shift 2

In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is 40 cm. The balancing length, if the radius of the wire AB is doubled, will be.....cm.

Space for your notes:



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Q16 - 28 July - Shift 1

A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length.

The ratio of new resistance to original resistance is:

- (A) 9 : 1 (B) 1 : 9
(C) 4 : 1 (D) 3 : 1

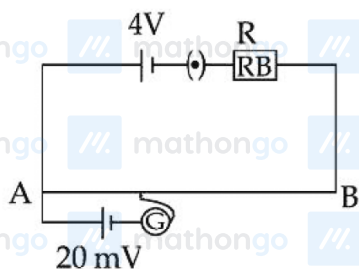
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Q17 - 28 July - Shift 1

As shown in the figure, a potentiometer wire of resistance 20Ω and length 300 cm is connected with resistance box (R.B.) and a standard cell of emf 4 V. For a resistance 'R' of resistance box introduced into the circuit, the null point for a cell of 20 mV is found to be 60 cm. The value of 'R' is

Space for your notes:

_____ Ω .



Q18 - 28 July - Shift 2

Given below are two statements :

Space for your notes:

Statement I : A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω .

Statement II : Two resistance $2R$ and $3R$ are connected in parallel in a electric circuit. The value of thermal energy developed in $3R$ and $2R$ will be in the ratio $3 : 2$.

In the light of the above statements, choose the most appropriate answer from the options given

below

(A) Both statement I and statement II are correct

(B) Both statement I and statement II are incorrect

(C) Statement I is correct but statement II is incorrect

(D) Statement I is incorrect but statement II is correct.

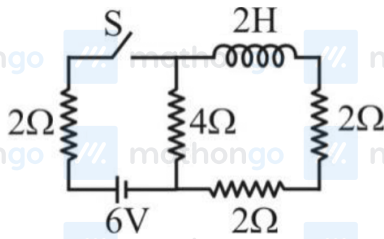
Q19 - 28 July - Shift 2

An electrical bulb rated 220 V , 100 W , is connected in series with another bulb rated 220 V , 60 W . If the voltage across combination is 220 V , the power consumed by the 100 W bulb will be about W .

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Q20 - 28 July - Shift 2

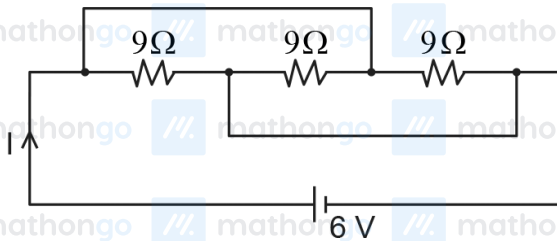
For the given circuit the current through battery of 6 V just after closing the switch 'S' will be



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Q21 - 29 July - Shift 1

The current I flowing through the given circuit will be _____ A.



Space for your notes:

Q22 - 29 July - Shift 2

Questions

MathonGo

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: Alloys such as constantan and manganin are used in making standard resistance coils.

Reason R: Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Q23 - 29 July - Shift 2

A 1 m long wire is broken into two unequal parts X and Y. The X part of the wire is stretched into another wire W. Length of W is twice the length of X and the resistance of W is twice that of Y. Find the ratio of length of X and Y.

- (A) 1 : 4
- (B) 1 : 2
- (C) 4 : 1
- (D) 2 : 1

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Space for your notes:

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Answer Key

Q1 (A)

Q2 (4)

Q3 (4)

Q4 (D)

Q5 (D)

Q6 (18)

Q7 (A)

Q8 (20)

Q9 (A)

Q10 (20)

Q11 (A)

Q12 (B)

Q13 (136)

Q14 (B)

Q15 (40)

Q16 (A)

Q17 (780)

Q18 (C)

Q19 (14)

Q20 (1)

Q21 (2)

Q22 (A)

Q23 (B)

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Q1 (A)

Electric displacement

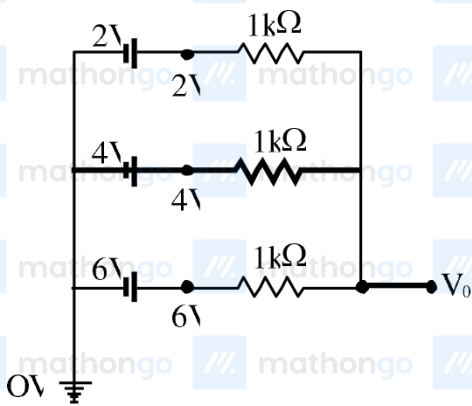
$$\vec{D} = \epsilon_0 \vec{E}$$

$$[D] = [\epsilon_0 E] = \left[\epsilon_0 \frac{\sigma}{\epsilon_0} \right]$$

$$[D] = [\sigma]$$

→ Surface charge density = σ .

Q2 (4)



By nodal analysis $\frac{V_0 - 2}{1k\Omega} + \frac{V_0 - 4}{1k\Omega} + \frac{V_0 - 6}{1k\Omega} = 0$

$$3V_0 - 12 = 0$$

$$V_0 = 4$$

Q3 (4)

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Each wire has resistance $= \rho \frac{4\ell}{\pi d^2} = r$

Eight wire in parallel, then equivalent resistance is

$$\frac{r}{8} = \frac{\rho \ell}{2\pi d^2}$$

Single copper wire of length 2ℓ has resistance

$$R = \rho \frac{2\ell \times 4}{\pi d_1^2} = \frac{\rho \ell}{2\pi d^2}$$

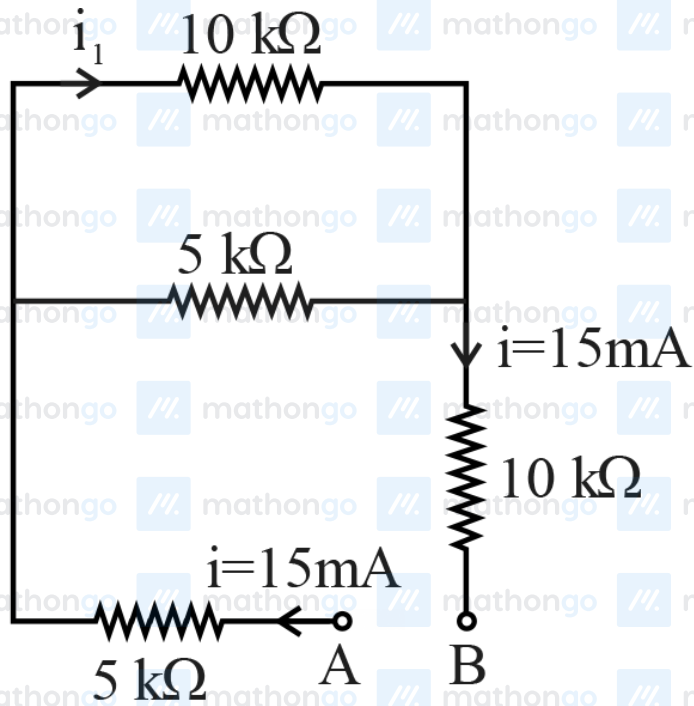
$$\Rightarrow d_1 = 4d$$

Q4 (D)

$$\text{modulation index} = \frac{V_{\max} - V_{\min.}}{V_{\max} + V_{\min.}} \times 100\%$$

$$= \frac{6 - 2}{6 + 2} \times 100\% = 50\%$$

Q5 (D)



$$i_1 = \frac{5}{10+5} \times 15\text{mA} = 5\text{mA}$$

$$V_A - 5i - 10i_1 - 10i = V_B$$

$$V_A - V_B = 75 + 50 + 150 = 275\text{ V}$$

Q6 (18)

$$1.2 = (\text{Potential Gradient}) \times 36$$

$$1.8 = (\text{Potential Gradient}) \times x$$

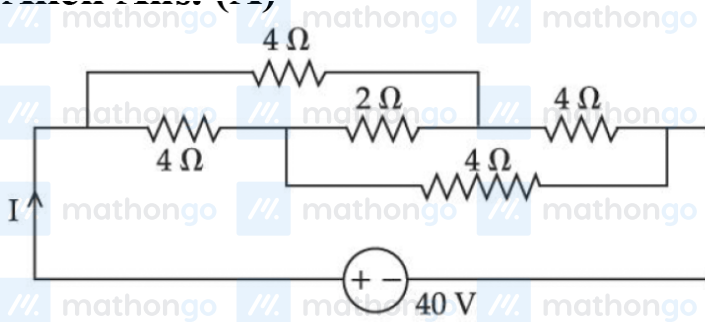
On dividing, we get

$$\frac{2}{3} = \frac{36}{x}$$

$$x = 18 \times 3 = 54\text{ cm}$$

$$\text{Hence difference} = 54 - 36 = 18\text{ cm}$$

Q7 (A)



Given circuit is balanced wheat stone bridge

Hence 2Ω can be neglected

$$R_{\text{net}} = 4\Omega$$

$$I = \frac{40}{4}$$

$$I = 10\text{A}$$

Q8 (20)

Sol. Initially, $\frac{P}{Q} = \frac{40\text{cm}}{60\text{cm}} = \frac{2}{3} \dots (1)$

Finally, $\frac{P+x}{Q} = \frac{80\text{cm}}{20\text{cm}} = \frac{4}{1} \dots (2)$

Divide (2) by (1)

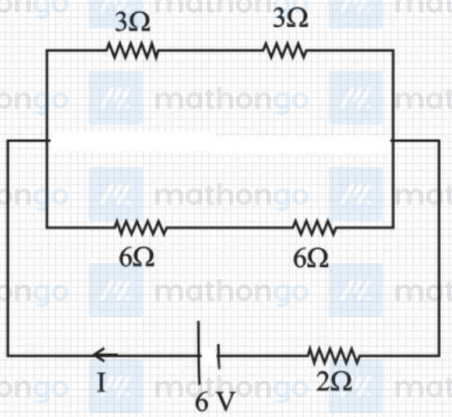
$$\frac{P+x}{P} = 4 \times \frac{3}{2} = 6$$

$$\Rightarrow 1 + \frac{x}{P} = 6 \Rightarrow \frac{x}{P} = 5$$

$$\therefore x = 5P = 5 \times 4 = 20\Omega$$

Q9 (A)

Balanced wheat stone bridge in circuit so there is no current in $5\ \Omega$ resistor so it can be removed from the circuit.



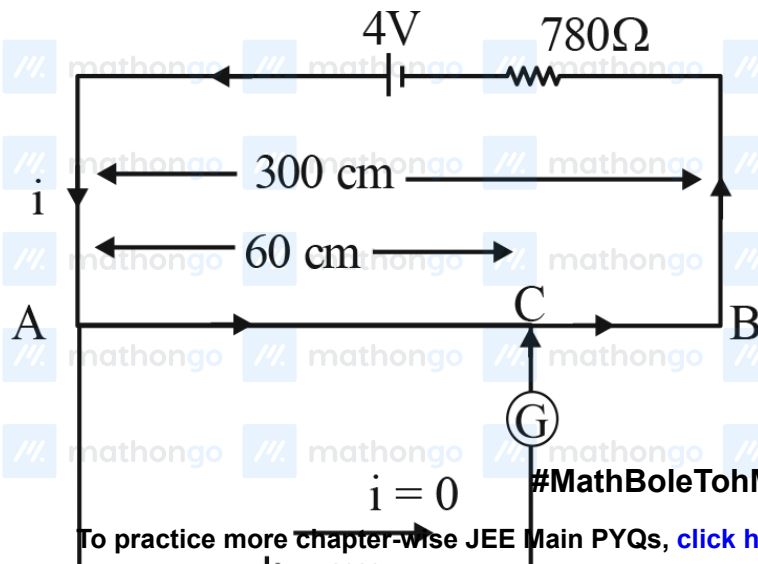
$$R_{eq} = \frac{6 \times 12}{6 + 12} + 2$$

$$= \frac{6 \times 12}{18} + 2$$

$$R_{eq} = 6\ \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{6}{6} = 1\text{ Amp.}$$

Q10 (20)



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Let resistance of potentiometer's wire is R

$$i = \frac{4}{R + 780}$$

Potential difference across AB

$$= \frac{4R}{R + 780}$$

Potential difference across AC

$$= \frac{4R \times 60}{(R + 780) \times 300} = \frac{4R}{5(R + 780)}$$

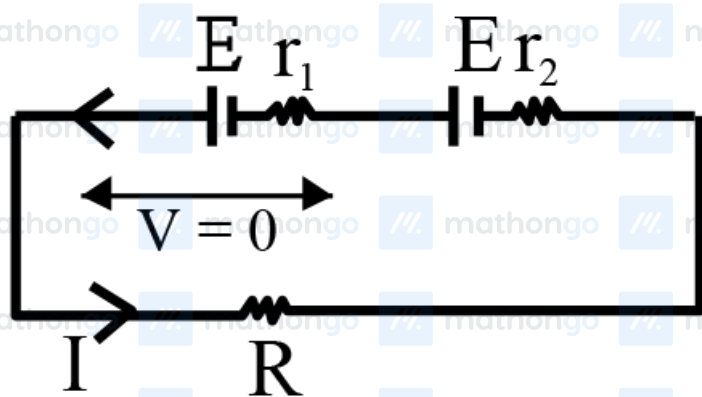
This should be equal to 20 mV

$$\frac{4R}{5(R + 780)} = 20 \times 10^{-3} = 2 \times 10^{-2}$$

$$4R = 10^{-1}(R + 780)$$

$$4R = \frac{R}{10} + 78$$

Q11 (A)



$$I = \frac{2E}{r_1 + r_2 + R}$$

$$IR = E - Ir_2$$

$$I(R + r_2) = E$$

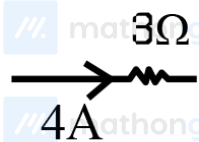
$$I = \frac{E}{R + r_2}$$

$$\frac{2E}{r_1 + r_2 + R} = \frac{E}{R + r_2}$$

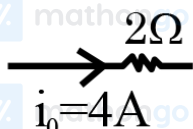
$$2R + 2r_2 = r_1 + r_2 + R$$

$$R = r_1 - r_2$$

Q12 (B)



DC



AC

$$H_1 = i^2 R_1 t \quad H_2 = i_{\text{rms}}^2 R_2 t \left\{ i_{\text{rms}} = \frac{i_0}{\sqrt{2}} \right\}$$

$$H_1 = 16(3)t \quad H_2 = \frac{i_0^2}{2} R_2 t$$

$$H_2 = 16t$$

$$H_1 : H_2 = 3 : 1$$

Q13 (136)

$$l = 1\text{m}$$

$$i = 1\text{A}$$

$$\text{Area} = 2 \times 10^{-6}$$

$$\rho = 1.7 \times 10^{-8}$$

$$R = \frac{\rho l}{A} = \frac{1.7 \times 10^{-8} \times 1}{2 \times 10^{-5}} = \frac{1.7}{2} \times 10^{-2}$$

$$V = \frac{1.7}{2} \times 10^{-2}$$

$$F = 1.6 \times 10^{-19} \times \frac{1.7}{2} \times 10^{-2}$$

$$= 1.36 \times 10^{-21}$$

$$= 136 \times 10^{-23}$$

Q14 (B)

$$\text{Drift velocity} = \left(\frac{e\tau}{m} \right) E$$

$$v_d = \left(\frac{e\tau}{m} \right) \left(\frac{\Delta V}{\ell} \right)$$

ΔV = Potential difference applied across the wire

As temperature increases, relaxation time decreases, hence V_d decreases.

As per formula, $V_d \propto \frac{1}{\ell}$

$v_d = \frac{I}{neA}$, as it is not mentioned that current is at steady state neither it is mentioned that n is constant for given conductor. So it can't be said that v_d is inversely proportional to A .

$$I = neAv_d = \frac{V}{R} = \frac{V}{\rho \ell} A$$

$$v_d = \frac{V}{\rho \ell ne} \quad \left(E = \frac{V}{\ell} \right)$$

$$v_d = \frac{eE\tau}{m}$$

τ decrease with temperature increase.

First and fourth statements are correct.

Q15 (40)

Independent of area in case of uniform wire.

Q16 (A)

$$R_1 = \rho \frac{L_1}{A_1}$$

$$R_2 = \rho \left(\frac{3L_1}{A_1 / 3} \right) = 9\rho \frac{L_1}{A_1}$$

$$\therefore \frac{R_2}{R_1} = 9$$

Q17 (780)

$$E = \frac{AC}{AB} (V_A - V_B)$$

$$\therefore 20 \times 10^{-3} = \frac{60}{300} \times \frac{4 \times 20}{R + 20}$$

$$\therefore R = \boxed{780} \, \Omega$$

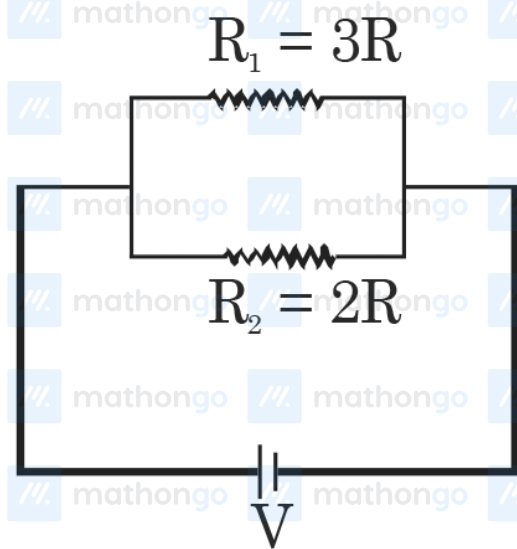
Q18 (C)

Statement 1 - $R = 80\Omega$

$$R_1 = R_2 = R_3 = R_4 = 20\Omega$$

In parallel $R_{eq} = \frac{20}{4} = 5\Omega$

Statement 2 –



$$P_{th} = \frac{V^2}{R}$$

$$\frac{P_1}{P_2} = \left(\frac{R_2}{R_1} \right) = \frac{2}{3} \text{ (where P is power)}$$

Q19 (14)

$$R_1 = \frac{V^2}{P} = \frac{220^2}{100} = 484$$

$$R_2 = \frac{V^2}{P} = \frac{220^2}{60} = 484 \left(\frac{10}{6} \right)$$

$$I = \frac{220}{484 + 484 \times \frac{10}{6}}$$

$$P_1 = I^2 R_1 = 14.06 \text{ W}$$

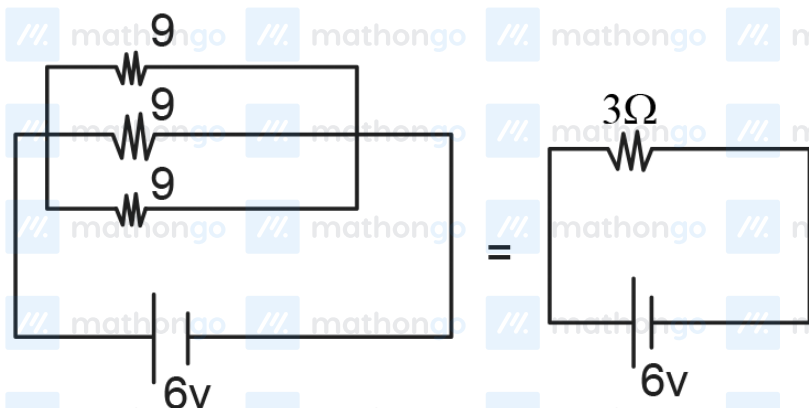
Q20 (1)

Just after closing the switch S, inductor behaves like an open circuit.

$$I = \frac{6}{2+4} = 1 \text{ A}$$

Q21 (2)

Sol. Equivalent circuit

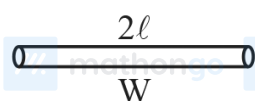
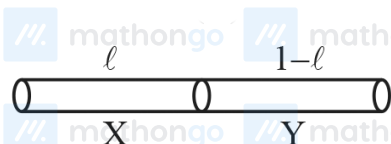


$$I = \frac{6}{3} = 2 \text{ A}$$

Q22 (A)

Theory based

Q23 (B)



$$\frac{R_X}{R_Y} = \frac{\ell_X}{\ell_Y}$$

When wire is stretched to double of its length, then resistance becomes 4 times

$$R_W = 4R_X = 2R_Y$$

$$\frac{R_X}{R_Y} = \frac{1}{2}$$

$$\text{So, } \frac{\ell_X}{\ell_Y} = \frac{1}{2}$$